When Does a Small Model Know to Hand Off? Reading Competence from Full-Duplex Speech Models for Real-Time Escalation

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Abstract

Full-duplex speech models answer in real time with a small on-device backbone, yet many queries exceed its competence. We ask whether the model's own hidden state signals, *before the first output token*, that a query is beyond it — reliably enough to hand off to a large cloud model without breaking the live session. Late-layer competence readouts turn brittle after duplex fine-tuning: final-layer probes collapse under distribution shift on text input (leave-one-pool-out math AUC .93→.37), consistent with late-layer specialization for duplex turn control, while verbalized self-assessment collapses on audio input. A mid-network linear probe survives both: leave-one-pool-out math .93, text-calibrated speech deployment .86, and an end-of-turn read in the streaming loop that reaches AUC .887 in $\sim$30 ms — before decoding starts. Run live end-to-end in a turn-based loop, the gated system climbs from .422 to .633 heard accuracy at 57% escalation, with the remaining gap dominated by the transcription uplink rather than the gate. The gate then transfers *frozen* to six public speech benchmarks using only label-free threshold quantiles, lifting Speech TriviaQA to .860 live against the model's official offline .755. A competence signal sufficient for real-time escalation is already present mid-network in duplex-trained speech models; the standard readouts simply miss it.

1 Introduction

A full-duplex speech model listens and speaks in the same forward pass, under a conversational deadline of a few hundred milliseconds [Stivers et al., 2009]. That deadline forces a small on-device backbone — and a small backbone routinely faces queries beyond its competence: long-tail facts, expert knowledge, multi-step reasoning. The natural architecture is selective escalation to a large cloud model, but the duplex setting imposes two constraints that existing routing work does not meet: the decision must come *before the first output token* (once the model starts speaking, a wrong answer is already on the air), and the session — audio context, voice, cache — must stay with the local model, which will speak the expert's answer itself.

This paper asks whether the small model already contains the signal this decision needs. We study MiniCPM-o 4.5 [Cui et al., 2026], a 9B omni/duplex fine-tune of Qwen3-8B, against matched-pair controls spanning raw backbones, streaming-omni, audio-only, and frozen-adapter fine-tunes, and evaluate every standard zero-shot competence readout — decode-time entropy, verbalized self-assessment ($p(\mathrm{True})$), and linear probes on hidden states — with judged failure labels on a frozen

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

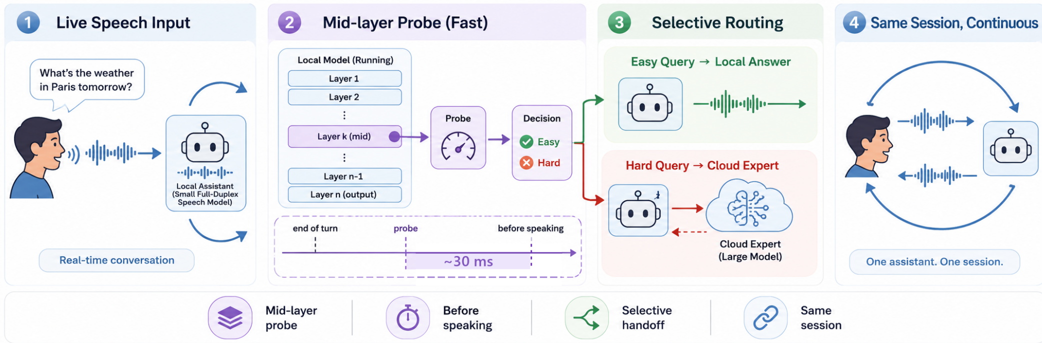

Figure 1: **(1)** Speech streams live into a small duplex-trained local model. **(2)** A mid-network (L22) linear read decides in $\sim$30 ms — before the first output token — so the expert call is already in flight while local decoding would just be starting. **(3)** Easy queries stay local; hard ones escalate to a cloud expert. **(4)** The *same talker speaks* the expert's answer in one continuous session, improving live spoken sessions (.422$\rightarrow$.633) and, frozen, public speech benchmarks (Speech TriviaQA .664$\rightarrow$.860).

pool of public-benchmark queries, in matched text and speech conditions. Throughout, *duplex* names the training regime, not the interaction protocol: we study duplex-trained weights driven through a *turn-based* loop whose only decision point is an end-of-turn read (§6); a deployed talker-on replay leaves the gate unchanged.

Everything runs on a frozen checkpoint — no fine-tuning; the only fitted component is a logistic regression on stored hidden states, calibrated once on text. Three findings, in increasing order of system-level consequence:

1. **Representation: duplex fine-tuning makes the late-layer readouts brittle; a mid-network signal survives.** The conventional final-layer probe looks strong in distribution (AUC .82) but is mostly a query-type shortcut: under leave-one-pool-out transfer it *inverts* on held-out math (.93$\rightarrow$.37). The damage is specific to the late-layer last-token position on text input — absent in raw backbones and non-duplex fine-tunes — consistent with late-layer specialization for duplex turn control. Mid-network (L22 of 36) the same linear read transfers at .93, and the mid-network optimum generalizes to a second, architecturally disjoint duplex family (hybrid-Mamba NemotronLabs-VoiceChat-11B).

2. **Modality: the mid-layer signal transfers text $\rightarrow$ speech in end-to-end models.** A probe calibrated on cheap text data reads speech-input hidden states at .86; the control that isolates the mechanism is Freeze-Omni, an adapter on frozen identical weights, where cross-modal transfer is dead ($\leq$.60). Verbalized self-assessment shows the mirror-image failure — it collapses on audio input — so the mid-layer probe is the one readout that survives both modalities. The transfer holds on real human recordings, not just TTS.

3. **System: the signal is available before the first token and works live, in and out of distribution.** The L22 read costs 20 ms (text) / 45 ms (audio) against time-to-first-token of 36/68 ms; in the streaming loop an end-of-turn read reaches AUC .887 in $\sim$30 ms. Run live end-to-end — expert consulted mid-conversation, answer injected, talker speaks it — heard accuracy climbs .422$\rightarrow$.633 at 57% escalation, and the frozen gate transfers to six public speech benchmarks with label-free thresholds, lifting Speech TriviaQA to .860 against the model's official offline .755.

Extended analyses — readout-damage controls, router baselines, escalation-channel diagnostics, latency mechanics — are in the appendix.

2 Related Work

Full-duplex real-time dialogue. End-to-end spoken dialogue spans native duplex designs — dGSLM [Nguyen et al., 2023], Moshi [Défossez et al., 2024], GLM-4-Voice [Zeng et al., 2024], single-LLM native full-duplex [Fang et al., 2026] — streaming omni models [Xu et al., 2025a, Cui et al., 2026], frozen-backbone adapters [Wang et al., 2024a], and audio fine-tunes [Chu et al., 2024]. A parallel line adds explicit turn-taking control: FlexDuo [Liao et al., 2025], MinMo [Chen et al., 2025], and Freeze-Omni's state heads decide *when the local model should speak*, at 120–250 ms cadences that show a live loop tolerates far more decision latency than our ∼30 ms read [cf. Riera et al., 2026, Wu et al., 2026]. Our gate decides the orthogonal question — *whether the local model should be the one answering* — and the head-based controllers' shared choice of a last-layer readout is exactly the point our results show duplex fine-tuning damages.

Does the model know what it knows? $p(\mathrm{True})$ self-evaluation [Kadavath et al., 2022], verbalized uncertainty [Lin et al., 2022], and truthfulness probes on hidden states [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024] established that correctness is readable internally; intermediate layers carry more of it than the output distribution [Orgad et al., 2025]. Semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024] needs multiple sampled generations; Semantic Entropy Probes [Kossen et al., 2024] amortize it into a hidden-state probe — the closest signal to ours, but read post-hoc on text-only half-duplex models. Multimodal uncertainty fusion [Zhang et al., 2026] asks a downstream question; we ask *where* in a duplex model a modality-robust read survives at all. Our contribution is a controlled account of when the standard readouts break under duplex fine-tuning — and where the surviving signal sits.

Selective escalation and routing. FrugalGPT [Chen et al., 2023], HybridLLM [Ding et al., 2024], RouteLLM [Ong et al., 2024], and confidence-based cascade serving [Xue et al., 2025] route from query-side features or preference data; RouterBench [Hu et al., 2024] and LLMRouterBench [Li et al., 2026] standardize evaluation and report that many routers barely beat simple baselines. AutoMix [Aggarwal et al., 2024] is the closest system design — a small model assesses its own answer and escalates — but the decision is *post-answer* self-verification; a voice deadline requires *pre-answer*. Both routing assumptions (query-side features; the large model answers the user directly) fail in full-duplex voice, where the session lives in the small model. Learning-to-defer [Madras et al., 2018], early exit [Schuster et al., 2022], speculative decoding [Leviathan et al., 2023], and test-time compute [DeepSeek-AI et al., 2025] make related decisions inside one model; our audit quantifies the query-side ceiling directly (§4.1).

The gap. Duplex work decides when to speak; routing work decides which model, from the query surface or after a first answer; uncertainty work asks whether the model is confident, almost always in text and offline. No prior work we know of asks whether a full-duplex speech model exposes a *pre-answer, modality-robust, internal* competence signal early enough to drive a live small-to-large handoff while preserving the session — the regime this paper measures end-to-end.

3 Experimental Setup

Target model and matched-pair controls. The system target is **MiniCPM-o 4.5** (9B; omni + full-duplex fine-tune of Qwen3-8B), bf16, greedy decoding, seed 42. To attribute effects to the *kind* of fine-tune rather than the backbone, we assemble ten models spanning raw backbones, a second duplex generation (MiniCPM-o 2.6), a streaming-omni fine-tune (Qwen2.5-Omni-7B), an audio-only fine-tune (Qwen2-Audio-7B), and a frozen-backbone adapter model (Freeze-Omni, LLM weights byte-identical to its raw control); the full roster is Table 4 (appendix). The comparison statistic is always the within-pair difference.

Query pool, labels, and audio arm. 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test): *trap* = SimpleQA [Wei et al., 2024] (the target fails 100%), *hard-*

Table 1: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix B.

signal	in-dist	LOPO math	audio input
max entropy@4	.696	≈chance on raw backbones	—
$p(\text{True})$-pre	.807	trap .945 (probe: .232)	**collapses**
$p(\text{True})$-post (full draft)	**.899**	—	degrades
final-layer probe	.822	**.372** (inverts)	.936 (intact)
mid-layer (L22) probe	.866	**.931**	**.855** (text-calibrated)

knowledge = MMLU-Pro [Wang et al., 2024b], *easy-fact* = TriviaQA [Joshi et al., 2017], *easy-chat* = Dolly/Alpaca [Conover et al., 2023, Taori et al., 2023], *hard-math* = GSM8K tail + MATH-500 [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]; public benchmarks only, no LLM-generated queries. Adequacy labels come from an LLM judge [Zheng et al., 2023] with structured outputs, blind to the gate and validated by a stronger re-judge (agreement .962 text / .945 audio; every headline number moves $\leq$.010 under the stronger labels); the escalation expert is `gpt-5.5` at low reasoning effort. The audio arm renders the same frozen pool to speech (single-voice TTS, content unchanged — only modality varies), following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]; audio-side findings are additionally validated on real human recordings (SD-QA, Faisal et al., 2021) and on the external benchmarks' official audio (§7). Judge prompts, pins, and spend: Appendix L–M.

Gate signals (no model training). (i) decode scalars: max token entropy / margin over the first k decode steps; (ii) linear probes: logistic regression on a single hidden state, position $\times$ layer swept, fit on the calibration split only; (iii) verbalized self-evaluation [Kadavath et al., 2022]: $p(\text{True})$-pre asks "would you answer this correctly?" before answering, $p(\text{True})$-post asks about a drafted answer; $P(\text{Yes})$ is read from the first token's distribution.

4 Reading Competence from Duplex Models

4.1 In distribution every readout looks fine; transfer reveals the shortcut

Table 1 compares the candidate signals on MiniCPM-o 4.5. In distribution the final-layer probe reaches .822 — but that number is mostly a *query-type* shortcut: the same hidden state classifies a query's source pool at 95.8%, a pool-identity oracle alone reaches AUC .715, and appending true pool labels to the probe adds nothing (.822→.821). The decisive test is leave-one-pool-out (LOPO) transfer, which we treat as the primary metric throughout: held-out math *inverts* to .372, and the mean escalation score on a 100%-fail trap pool is .232. An external query-surface router makes the same point from the other side: under identical labels a TF-IDF router reaches only OOF AUC .669 (below the pool oracle), collapses to .38–.57 under LOPO, and $100\times$ the data on a public routing benchmark buys back only .04 of AUC (Appendix D). Instance-level competence information lives in the model's internal state, not on the query surface.

Verbalized self-assessment transfers differently: *pre-answer* $p(\text{True})$ catches the trap pool the probe misses (escalation score .945 vs. the probe's .232) — but timing is essential: asked *after* drafting, the model endorses its own confident-wrong answer (.360). Post-draft $p(\text{True})$ is the strongest single signal (.899) but requires paying for a full draft, which a voice deadline cannot afford.

4.2 The damage is duplex-specific; the signal survives mid-network

Matched pairs attribute the probe failure to the *kind* of fine-tune, not the backbone: the raw Qwen backbone, label-matched to the duplex model's fail rates, holds LOPO math at .825 where the duplex model sits at .372; streaming-omni and audio-only fine-tunes are undamaged; a second duplex gen-

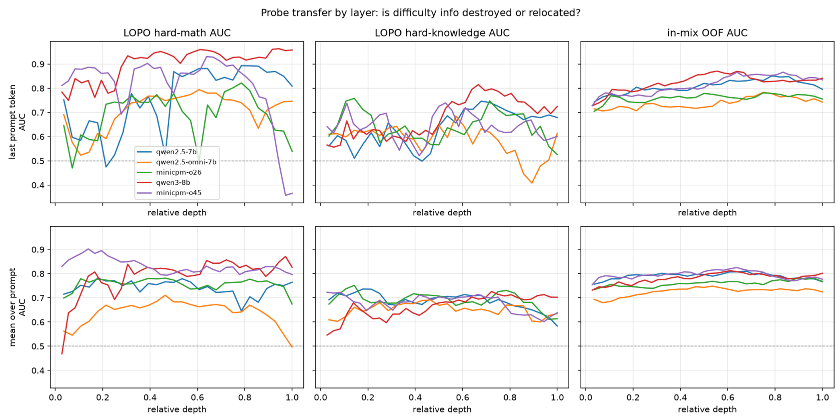

Figure 2: The core representational result: LOPO math AUC by layer and position. On the duplex model the standard (final-layer, last-token) read inverts on text input while mid-network layers hold $\geq .9$; raw backbones show no cliff. The deployed gate reads L22.

eration (MiniCPM-o 2.6) reproduces the collapse (Appendix B). A layer $\times$ position sweep localizes it (Figure 2): the collapse is specific to (late layer $\times$ last token) on text input — mean-pooled reads at the same depth are fine, and mid-network the last-token read recovers to **.931 at L22**, near the raw backbone. This is the position a streaming head must re-purpose to encode turn-control state, and the pattern is consistent with late-layer specialization for duplex turn control overwriting the read-out while the underlying information survives. The mid-network optimum is not a MiniCPM quirk: on NemotronLabs-VoiceChat-11B — a hybrid-Mamba duplex model, architecturally disjoint — the same recipe peaks mid-band at L30–34 of 56 and transfers to the external pools at AUC .70–.79 with a quarter of the calibration data (Appendix C).

4.3 Text-calibrated, speech-deployed — in end-to-end models only

Probes trained on text hidden states and scored on speech-input hidden states locate where the representation becomes modality-shared: early layers are modality-specific (.54–.60), and from $\sim$50% depth text$\rightarrow$audio transfer plateaus at .82–.87 (.855 at the deployed L22). This replicates on MiniCPM-o 2.6 and cross-family on Qwen2.5-Omni. The decisive control is **Freeze-Omni**: identical frozen LLM weights with audio attached by an adapter — and cross-modal transfer is dead (.52–.60). The modality-shared mid-layer core is *created by training the backbone on the modality*; the cheap recipe "calibrate on text, deploy on speech" is valid for end-to-end models and invalid for adapter-attached ones. Verbalized introspection shows the mirror-image failure: on audio input, pre-answer $p(\mathrm{True})$ collapses on both MiniCPM duplex generations while both non-duplex controls stay intact; controlled arms show the verbal self-check runs over text-token pathways that audio embeddings do not feed — duplicating the question as text restores it — while a linear probe reads the same information off the same forward pass at .93 (Appendix B). All audio-side findings replicate on 200 real human recordings (SD-QA), ruling out TTS artifacts. **The mid-layer probe is the one readout that survives every training regime that can converse**; the live system deploys the same L22 read plus two zero-latency pooled positions (the *deployed* probe; Appendix B).

5 From Signal to Gate

The decision exists before decoding starts. The gate is one 4096-d dot product on a hidden state the forward pass already computes. Measured on H100, the L22 decision lands in **20 ms (text) / 45 ms (audio)** against time-to-first-token of 36/68 ms: $T_{\text{gate}} < T_{\text{first token}}$, far inside the 200–300 ms turn-taking budget [Stivers et al., 2009], and the cloud call can launch while the remaining $\sim$40%

Table 2: Deployment policies, frozen test split (n=240); latencies measured.

policy	acc	lat. mean (s)	P50	P95
fast-only	.588	3.0	3.5	4.2
all-THINK (built-in)	.637	22.4	17.2	60.1
gated-think @.33	.613	12.1	3.6	47.6
gated-cloud @.15	.688	5.3	3.6	10.1
gated-cloud @.33	**.787**	**6.5**	3.6	20.8
gated-cloud @.50	.858	7.0	3.6	24.4

of prefill and the whole decode still run ($p(\text{True})$-pre costs an extra prefill, $p(\text{True})$-post a full draft: 1.9/3.5 s; per-layer cost curves in Appendix E).

Accuracy–cost tradeoff. On the frozen test split (n=240): small-only .588, big-only (`gpt-5.5`) .917. Gated escalation beats random escalation at every operating point (Figure 6, appendix); the balanced tier at a 33% budget reaches **.779–.787**, recovering ∼58% of the small→big gap at one third of the big-only cost. In distribution the gate is statistically on par with a pool-oracle (type-only) router — as the type-shortcut audit predicts — so its case over type routing rests on transfer, where the type shortcut collapses (§4.1) and the mid-layer signal does not.

Against the model's own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode — self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency at every escalation rate (gated-cloud@.33: .787 at mean 6.5 s vs. all-THINK .637 at 22.4 s; Table 2). The reason is mechanistic: the gate predominantly flags knowledge and trap failures, which extra compute cannot fix — thinking helps math and hurts chat — so external escalation is necessary, not merely better.

6 Live End-of-Turn Escalation

The streaming loop. The protocol is turn-based: audio streams in 1 s chunks, but the talker speaks only after the user's turn ends and the model's native listen/speak head is not engaged. Per-chunk probe scores are strictly weaker than the offline read (best chunk statistic AUC .764) and miss short utterances whose critical entity lands in the final chunk. The fix is an **end-of-turn read**: after the last audio chunk, prefill an assistant turn containing a single space and read L22 at the assistant-start position. This read reaches **AUC .887** — above the offline reference (.843) — in ∼30 ms, still before the first output token, and is the loop's only decision point; escalation-rate thresholds are quantiles of this single score. Speculative mid-stream prefetch was measured and dropped (an expert answer to a partial transcript answers the full question only .07–.51 of the time; Appendix F). On escalation the talker speaks a stall phrase, its *own* transcription of the audio goes to the expert (`gpt-5.5`, reasoning effort fixed low — higher effort buys zero accuracy on the escalated population at double the latency), and the returned answer is injected as context for the talker to relay. On the deployed channel the relay is clean: model wrong, correct answer injected, comply $= 1.00$ (n=24); more broadly the competence signal also predicts relay compliance (comply .42→.75 across gate-score quartiles) — a model that knows it doesn't know is willing to listen. Conflicting-injection and framing controls: Appendix F.

The live result. We ran 240 frozen test queries × four escalation tiers, each a complete turn-based spoken session, scored on *heard accuracy* — what a user would actually have heard. Headline numbers use a pre-registered input-side filter (21 formula-LaTeX ids + 1 broken recording removed; spoken formulas are destroyed by any TTS render before the system hears them). Heard accuracy climbs monotonically, $.\mathbf{422} \rightarrow .\mathbf{528} \rightarrow .\mathbf{592} \rightarrow .\mathbf{633}$ at 0/14/31/57% escalation, every tier's gain significant (paired bootstrap; Figure 3). The remaining gap to the channel-controlled counterfactual (escalated rows re-scored with gold-text expert answers) is dominated by *transcription-uplink* errors, not gate failure: the channel costs +.023/+.069/+.128 across tiers, and in the channel-controlled view

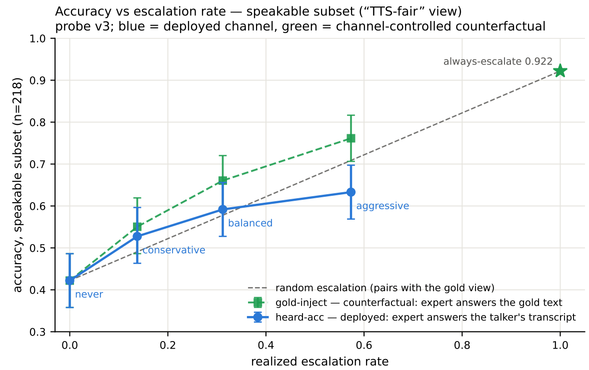

Figure 3: Live dual-view tradeoff (n=218 speakable subset). Blue: honest heard accuracy of the live system (paired-bootstrap CIs). Green: the channel-controlled counterfactual (same sessions, escalated rows re-scored with gold-text expert answers) with its paired random reference (dashed). The view-to-view gap is the speech-channel price; the floor-to-floor gap to offline operation is the audio + streaming-answer tax.

the gate clears its paired random reference at every arm. Live arm accuracies on this pool carry a replication-noise floor of $\pm$.02–.03, so no claim here rests on a sub-3-point live delta. Full tables (both views, both pools), latency profiles, uplink diagnostics, and two boundary query classes (false-premise, real-time) are in Appendix F. A talker-on replay (237/239 turns barge-in-seeded) leaves the frozen probe intact: AUC .871/.856 vs. .866 headless (Appendix G). A TTS-on re-run of the balanced arm puts first *audible* audio at p50 .55 s / p99 .64 s after end of turn on local answers and 22–29 ms on escalated ones — the canned stall starts speaking while the expert call is in flight (Appendix F).

7 Frozen Transfer to Public Speech Benchmarks

The out-of-distribution test the type-shortcut audit demands: we take the deployed gate *frozen* — weights, features, and layer choice fit once on the internal calibration mix — and run the complete live loop of §6 on six public speech benchmarks: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] (spoken renderings of Joshi et al., 2017, Berant et al., 2013, and the set of Nachmani et al., 2024), SD-QA (real dialectal human speech) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All audio is the benchmarks' official pre-rendered audio; no query ever touches probe calibration. The only per-pool adaptation is *label-free*: tier thresholds are quantiles of the probe's own score distribution on each pool, consuming no labels and no judge calls. It buys budget control, not discrimination: one absolute threshold frozen on the internal pool still beats matched-rate random on four of five pools, but fires at 2–48% across them (Appendix H). Where an official protocol exists we score with it (the OpenAudioBench judge; VoiceBench's 1–5 judge), and our chat-mode control reproduces the official anchors on-scale.

The frozen gate improves all five QA benchmarks (Table 3): thresholds fire at sane rates, every curve rises monotonically, and 50% escalation lifts the live floor by +.11 to +.29. On Speech TriviaQA the 9B system reaches **.860 live** — against its own official offline .755 and .629 for Qwen3-Omni-30B [Xu et al., 2025b] at 3.3$\times$ the parameters: selective cloud escalation outperforms larger standalone backbones. Transfer AUC holds a .68–.81 band on every pool against .879 internally, and the cleanest evidence that the signal is difficulty rather than surface is cross-lingual: an English-only calibration expansion moved *Chinese* Reasoning QA from AUC .621 to .683, the largest per-pool gain. On Llama Questions — single-type, so no shortcut to exploit — **selective escalation matches the expert at half the expert calls**: the probe's split is decisive (kept-local .976 vs. escalated .696, z=6.46), the expert's whole advantage lands where the probe sends it (.696$\rightarrow$.888, p<.0001), and

Table 3: The frozen gate on six public speech benchmarks (live loop; realized escalation 15/30/50%). Judge: OAB / VB = official; ours = reference-anchored. Ceiling = GPT-5.5 on gold text, same judge; official = MiniCPM-o 4.5's reported offline number. Live accuracies carry a $\pm$.02–.03 replication floor.

pool	judge	n	escalation tier 0%	15%	30%	50%	ceiling	AUC	official
Speech TriviaQA	OAB	250	.664	.732	.800	**.860**	.968	.789	.755
Speech Web Questions	OAB	250	.57	.628	.680	.732	.864	.785	.702
Llama Questions	OAB	250	.84	.904	.904	.948	.928	.806	—
SD-QA	ours	200	.51	.600	.720	.795	.930	.792	—
Reasoning QA (zh)	ours	202	.59	.653	.713	.772	.871	.683	—
AlpacaEval (1–5)	VB	199	3.99	3.96	4.23	4.35	4.96	—	4.8

always-escalate buys nothing measurable over selective; matched-rate precision reaches 95% of the base-rate cap (Appendix I).

Two honest negatives. **AlpacaEval**: the gate does not beat random — open-ended instruction following has no "missing fact" event for a retrieval-failure detector to read. **Web Questions and SD-QA are flat** under the deployed-probe refit: on Web Questions strict alias-based labels decouple "wrong" from "uncertain" (even the gold-text ceiling is .864), and the failures the gate ranks highest are ones the expert misses too (expert-fixable share .66 at the 15% cut vs. .75 pool-wide). The external channel cost is small (heard-vs-gold gap .035–.056 at aggressive vs. .175 internally), and SD-QA's escalated subset converts at .82–.84 on real dialectal speech. Routing can also *reduce* tail latency, replacing slow wrong local decodes with short expert round-trips (Reasoning QA P90 13.3→10.9 s; Appendix I).

8 Discussion

What the signal detects. The pre-answer probe is specifically a *retrieval-failure* detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is an observable event in the model's state. Execution failures (getting stuck mid-derivation) surface in *decode-time* signals instead, and metacognitive failures — questions with false premises, where no failure event occurs in the model's experience — are invisible to every signal we tested, query-surface routers included (fire rate .18 vs. trap's .90; Appendix F). This scope motivates a decode-time check behind the pre-answer gate.

Limitations. (i) One phrasing per $p(\mathrm{True})$ variant. (ii) Real-speech validation covers scripted read speech in one dialect. (iii) Formula-symbolic content is out of scope for the audio arm — destroyed by any TTS render, an input-side limit distinct from the transcription-uplink loss. (iv) The second-family replication covers mid-band readability and frozen-recipe transfer, not the full matched-pair attribution or the live loop. (v) Labels come from one judge family, bounded by a stronger re-judge (deltas $\leq$.010), without human labels. (vi) The live sweep is one session per query per tier ($\pm$.02–.03 replication floor); the talker-on replay (Appendix G) leaves concurrent-with-generation reads open.

9 Conclusion

Duplex-trained speech models carry a competence signal their standard readouts miss: a mid-network linear read that survives the readout damage, transfers from text calibration to speech, and lands tens of milliseconds before the first output token. Deployed live, it turns a 9B talker into a system that knows when to hand off; frozen, it lifts public speech benchmarks past models three times its size. Reading the right layer makes real-time escalation possible.

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A Model roster

Table 4: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

B Extended signal and readout analyses

ROC curves. Figure 4 shows the calibration-split ROC for the signals of Table 1.

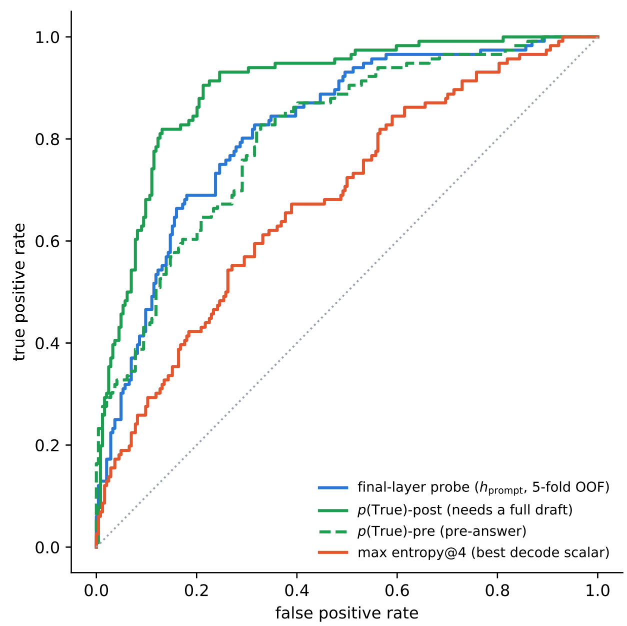


Figure 4: ROC on the calibration split. $p(\text{True})$-post leads but needs a full drafted answer; the probes and $p(\text{True})$-pre are pre-answer.

$p(\text{True})$ **backbone dependence.** Across backbones, post-draft $p(\text{True})$ beats the trained final-layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, .685 vs. probe .798) shows the pattern depends on the backbone's self-evaluation quality.

Matched pairs. Table 5: with label coverage matched by subsampling the raw control to the duplex model's per-pool fail rates, the raw backbone keeps LOPO math at .825/.809 while duplex models sit at chance — label coverage is ruled out; the representation changed. Degradation is ordered raw > omni-streaming > duplex. Table 6 gives the by-depth summary behind Figure 2: the within-pair deficit at the best mid-layer is only $-.03$ vs. $-.59$ at the final layer; mean-pooled probes survive to the final layer on duplex models; duplex damage is severe but local, omni-streaming damage mild but diffuse.

Table 5: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	**0.809**	0.745	**0.538** $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	**0.526** $\approx$ chance	0.61

Table 6: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	**0.931 (L22/36)**	**0.366**	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	**0.822 (L21/28)**	**0.540**	cliff

Modality specificity of the cliff. Under audio input the final layer does *not* invert (L35 audio LOPO math .936 vs. .366 text); a speak-mode template control on text input leaves the cliff unchanged (.362) — the operative variable is the modality of the input context, not the output mode.

Audio collapse of verbalized introspection. Table 7. Three controlled arms pin the mechanism on MiniCPM-o 4.5: *not perception* (ASR audit WER .074; $p(\mathrm{True})$ on the model's own transcript snaps back to .074; the well-heard subset still collapses); *not a context prior* (irrelevant filler audio does not inflate p_{yes}); *binding* (duplicating the question as text alongside the audio fully restores introspection, .034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at .93+. The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking ("repeat-then-judge").

Table 7: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes}: text $\rightarrow$ audio	audio $p(\mathrm{True})$-pre AUC
MiniCPM-o 4.5	duplex	.055 $\rightarrow$ **.556** (collapse)	.786 (trap dead)
MiniCPM-o 2.6	duplex	.196 $\rightarrow$ .345	**.491** $\approx$ chance
Qwen2.5-Omni	omni-streaming	.279 $\rightarrow$ .213 (intact)	.727
Freeze-Omni	frozen + adapter	.165 $\rightarrow$ .112 (intact)	.612 (capability caveat)

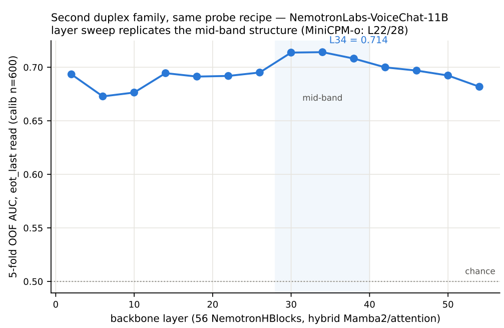
Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (.400$\rightarrow$.450), audio overconfidence (paired p_{yes} shift $+.089$; failure subset .415$\rightarrow$.581), and the layer structure (text-calibrated read on real-speech audio holds .76–.80 at L22–L26, peak .800 at L25 — transfer across both content and modality).

Synthesis. End-to-end multimodal training *builds* the shared mid-layer core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

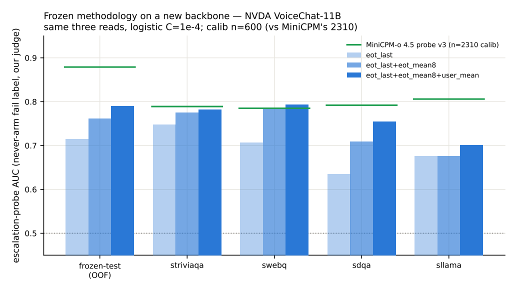
The deployed probe. The mechanistic probe is a single-position L22 read. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean), refit on a calibration mix widened to 2,310 public-benchmark queries disjoint from every evaluation benchmark; a pre-registered guard held (frozen internal test AUC .860→.879) and feature selection never saw the externals. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe's in-distribution area advantage is modest — its real advantage is LOPO robustness, which an in-mix test cannot show.

C Second duplex family: NemotronLabs-VoiceChat-11B

The pre-registered prediction test: with the identical recipe (same audio, same judge, 600-query calibration), the layer sweep on this hybrid-Mamba duplex model (56 blocks: 27 Mamba-2 / 4 attention) peaks mid-band at L30–34 (eot-last AUC .714 vs. .693/.682 at the endpoints); stacking the same three positions lifts OOF .714→.761→.790; the frozen-methodology probe transfers to the external pools at AUC .781/.793/.754/.701 — the deployed-probe band reached with a quarter of the calibration data. Boundary notes: the backbone is much weaker at knowledge retrieval (striviaqa local .32 vs. MiniCPM .62, same judge) and English-only (the cross-lingual pool has no analog). The live-loop port and the trained-checkpoint ablation remain future work.



(a) layer sweep



(b) external transfer

Figure 5: Second-family test: (a) the usable read again peaks mid-network; (b) frozen-recipe transfer lands in the deployed-probe AUC band.

D Query-surface router audit

A RouteLLM-style TF-IDF→LR router under identical labels reaches OOF AUC .669 (below the pool oracle's .715); a 24-configuration sweep tops out at .689, so this is the query surface's ceiling, not tuning. Under LOPO it collapses to .38–.57 and scores the 100%-fail trap pool at .230 — it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 7). Nor is it data-starved: the identical recipe on RouterBench [Hu et al., 2024] (36,497 prompts) reaches only in-domain AUC .710 with leave-one-benchmark-out at chance; RouteLLM's released preference-trained routers score .523/.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (.440). On audio labels the router improves (test .814) but still trails the mid-layer probe (.879) and cannot fire mid-stream. In distribution, the gate's area over random ($+.054$) is statistically on par with the pool oracle's ($+.042$; paired bootstrap delta CI $[-.003, +.027]$, p=.13) — the gate's case is transfer, not the in-distribution margin. $p(\mathrm{True})$ tradeoff variants: Figure 8.

E Gate timing details

Per-layer truncated-forward cost vs. quality (Figure 9): the L22 hidden state is available at $\sim$57–61% of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at $\sim$55–60% depth. If the cloud call launches at

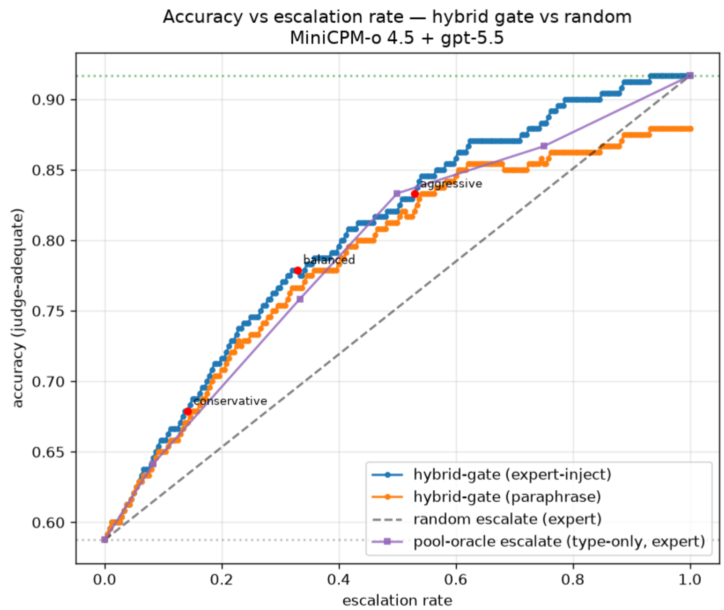


Figure 6: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

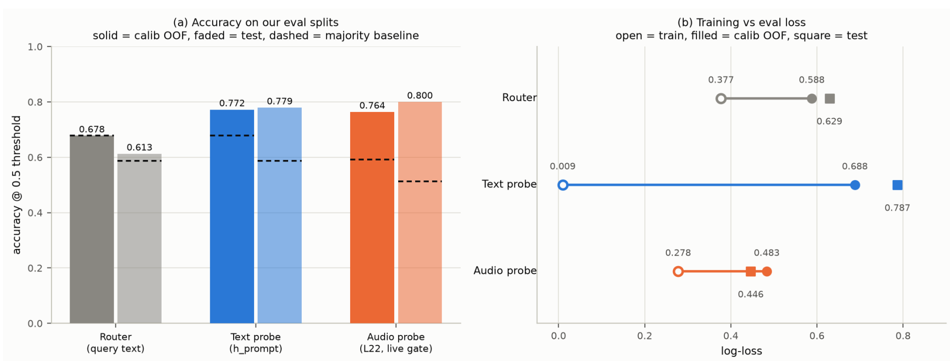


Figure 7: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but only $\sim$20–31% of *escalated* queries (they skew to short local answers); the residual wait is P50 2–3 s, P90 $\sim$27 s (Figure 10) — the stall-phrase budget the injection protocol must cover.

F Live-loop details

Full live tables. Table 8 (both scoring views, speakable subset and full pool) and Table 9 (measured response latency). The conservative tier *shortens* the P95 tail: its few escalations displace exactly the longest local decodes. Figure 11 joins accuracy and latency; Figure 12 replays three real sessions from recorded timers.

Time to first audible audio. The sweep above is text-out; a TTS-on re-run of the balanced arm (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker's own voice) measures speech onset directly. The gate decision is unchanged — the TTS token enters the context only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From

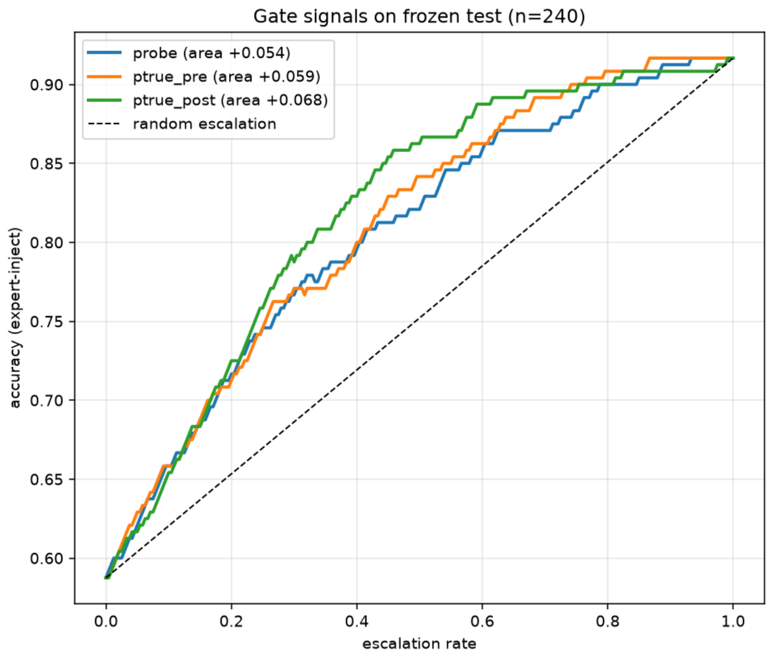


Figure 8: Offline tradeoff for the $p(\mathrm{True})$ signals (pre-answer and post-draft).

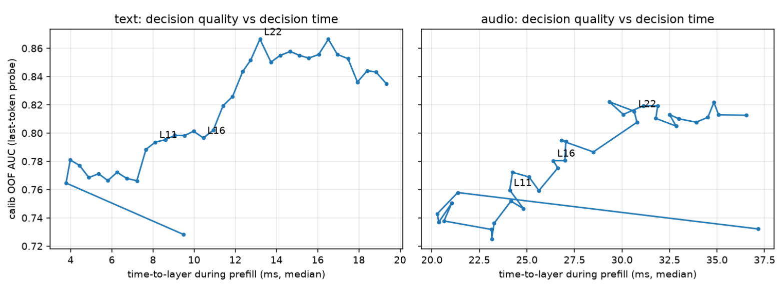


Figure 9: Fork-depth Pareto: per-layer wall-time vs. gate quality.

gate decision to first audible audio: **local answers p50 .552 s / p95 .599 / p99 .644**; **escalated turns p50 .022 s / p99 .029** — the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the escalated path reaches first audio 25× faster. Expert *content* becomes audible at p50 6.4 s / p95 26.0 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 .66 s after the answer returns); the stall covers the head of that wait, and Table 9's completion profile bounds the tail.

Speculative prefetch, measured and dropped. Framed as speculative escalation (draft–verify at sentence granularity), its acceptance rate — does an expert answer to the partial transcript answer the full question — is .07/.19/.51 at 25/50/75% of the audio; at the mid-stream gate's typical fire point 6–8 of 10 calls would be wasted. The trap pool's floor (.00/.10/.27) shares the back-loaded-core property that breaks mid-stream probe reads.

Conflicting-injection controls (172 sessions). Fabricated conflicting expert answers by within-pool derangement, four framings: worst-case comply is .00 neutral / .53 deployed authority / .79 strong-override, while *seeding* words into the talker's mouth backfires (comply .25, lip-service .62 — forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed. Model-wrong × wrong-injection swallow rate is .64: expert quality is the accuracy ceiling of the cascade.

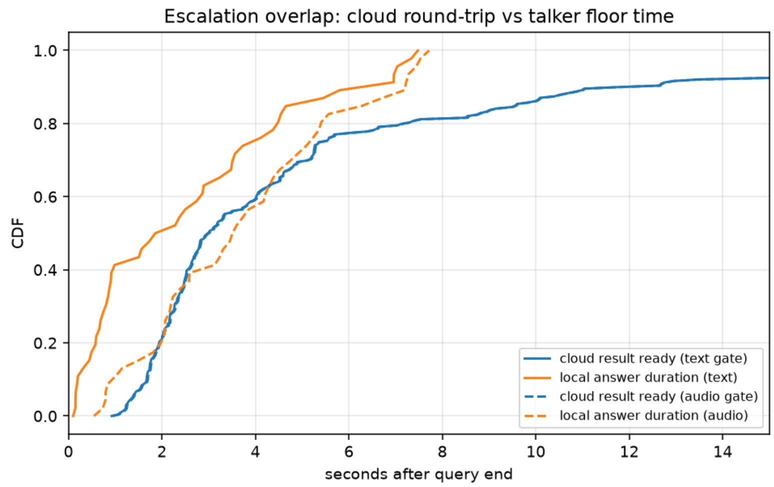


Figure 10: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

Table 8: Live streaming sweep, paired bootstrap 95% CIs (B=10^4). Top: speakable subset (n=218); bottom: full pool (n=240). Gold-inject = escalated rows re-scored with gold-text expert answers; channel cost = paired difference. * CI excludes 0; † CI lower bound at zero.

tier	esc	heard-acc	Δ vs. floor	gold-inject	channel cost
never	0%	.422 [.36,.49]	—	.422	—
conservative	14%	.528 [.46,.59]	+.106*	.550	+.023†
balanced	31%	.592 [.53,.66]	+.170*	.661	+.069*
aggressive	57%	.633 [.57,.70]	+.211*	.761	+.128*
always (synth)	100%	—	—	.922 [.89,.95]	—
never (full)	0%	.383 [.32,.45]	—	.383	—
conservative	16%	.483 [.42,.55]	+.100*	.525	+.042*
balanced	35%	.554 [.49,.62]	+.171*	.654	+.100*
aggressive	61%	.596 [.53,.66]	+.212*	.771	+.175*
always (synth)	100%	—	—	.917 [.88,.95]	—

Uplink diagnostics. On the escalated population the expert answering the talker's own transcript scores .585 vs. .871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap (.585→.694, McNemar p=.007) — diagnostic only (its critical-path latency and external-pool survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the audio channel itself is lossless for speakable content: the leak is the talker's transcription, not the synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated heard-failures, first-generation sweep).

Boundary query classes. *False premises*: the model fails substantially (.63 text / .47 audio) and the expert would help (.80), but the gate is blind (fire@balanced .18 vs. trap's .90; no score separation) — answering along a false premise does not feel like inability. *Real-time queries* ("what is NVDA trading at"): the gate is not blind, the class was simply absent from every static training family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori, never-changing as in-family controls) raises held-out fast-changing fire rates to .80/.93/1.0 across tiers with frozen-test AUC unchanged (.877 vs. .879). One calibration subtlety: tier budgets must remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold shift.

Table 9: Measured response latency (s), n=240 per arm (latency is channel-independent). P99 on n=240 is indicative only.

tier	mean	P50	P95	P99	Δmean	ΔP50	ΔP95	ΔP99
never	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

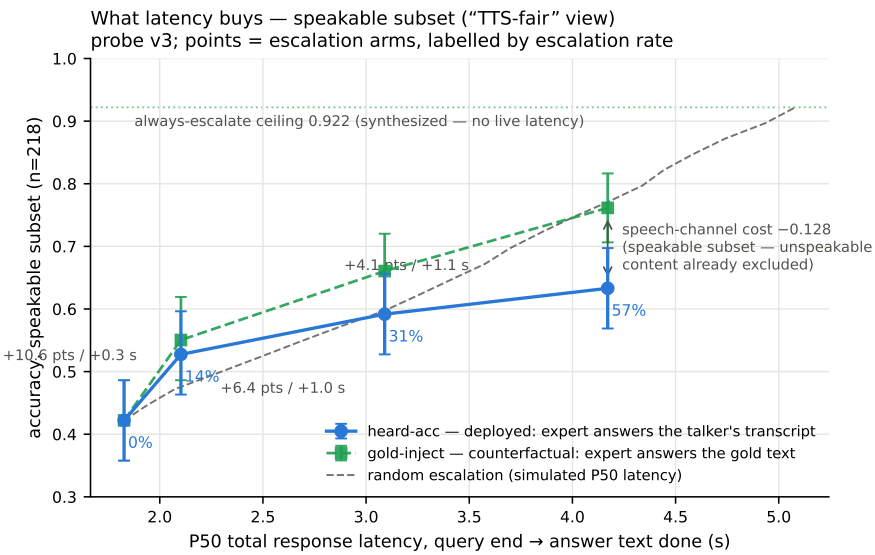


Figure 11: Accuracy vs. per-arm P50 latency (speakable subset), both scoring views, with the simulated-P50 random reference.

Demonstration system. The pipeline runs as an interactive web demo on one H100: live microphone audio, end-of-turn read (∼54 ms) against the balanced threshold, escalated turns stall, consult the expert, and *speak* the relayed answer through the model's native TTS head (probe context and thresholds untouched). Two demo-only extensions: a web-search tool for the expert (real-time queries then return live values), and backchannel-aware barge-in — speech over the talker ducks playback, a fast ASR pass returns the floor on backchannels (“ok”, “mm-hm”; an English+Chinese lexicon) or commits the interrupt, aborting generation at chunk granularity and seeding the next turn. All measured arms keep the tool-free expert and turn-based loop; the demo will be made available upon publication.

G Duplex-validation sweep: the gate under the talker

The headless live sweep (§6) never engages the talker head; this sweep replays its full 240-query pool through the deployed voice loop above to test whether the frozen probe read survives the spoken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation disabled (the arm isolates gate validity, not routing outcomes). **Clean**: the query streams as mic-shaped PCM (pool TTS audio plus steady noise at RMS .008), the energy VAD ends the turn, the talker answers aloud, and the session is cut after the first audible chunk. **Overlap**: an easy warmup question (probe score <.25, locally answered, >5 s spoken answer) puts the talker mid-answer, and the target query is then spoken over it; sustained overlapping speech commits a barge-in whose audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

Against the frozen local-failure labels of the same pool, the identical frozen probe reads: headless AUC .866 [.817, .909], clean .871 [.824, .914], overlap .856 [.803, .901]; barged-only subset

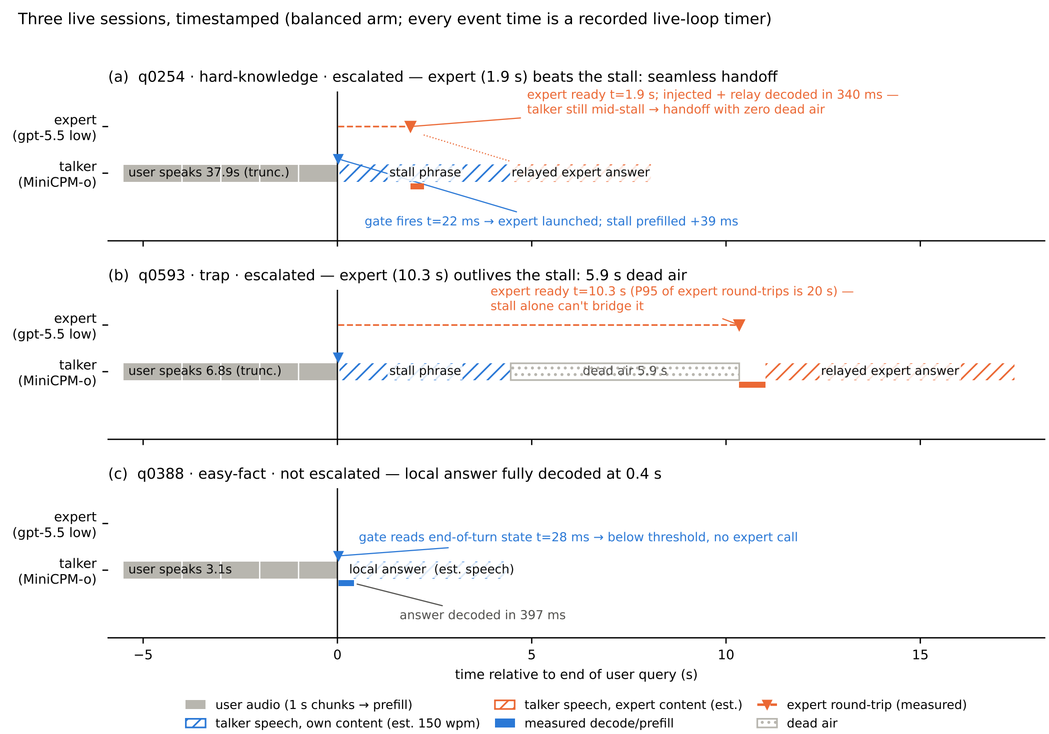


Figure 12: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

(n=237) .854. Paired bootstrap deltas vs. headless are $+.005$ $[-.025, +.035]$ (clean) and $-.009$ $[-.036, +.016]$ (overlap) — both inside the $\pm$.02–.03 live replication floor. Per-query scores correlate across regimes at r=.90–.92, and the balanced threshold makes the same fire/hold decision as the headless read on .887 (clean) and .912 (overlap) of queries. The end-of-turn read costs 24–26 ms at p50 (p95 $\leq$30 ms) in both arms, with the first audible answer chunk arriving $\geq$0.5 s after the gate decision (p50 $\approx$0.58 s), so the decision precedes any audible commitment. Scope: the loop keeps per-turn session control — the mic stays open while the talker speaks and interruptions seed the next read, but incoming audio is not prefilled *during* generation; validating a probe read concurrent with generation remains open.

H Fixed-threshold transfer and the expert-benefit oracle

Both analyses below run in the *gold-inject* view (escalated rows take the expert's answer to the gold text). The deployed heard view cannot evaluate a freely moving threshold: the most permissive arm escalates 50%, so the other half of each pool has no expert outcome on record, whereas the ceiling runs cover every id. Labels follow each pool's own judge (Table 3); the two judges disagree by up to 13 points on Web Questions, so they are never mixed within a row. AUC columns are recomputed from the deployed sweep's stored end-of-turn scores; they differ from Table 3's AUC column — captured in a separate scoring pass — by up to .03 externally and .04 on the internal pool (.840 here vs. .879 there): the passes render and prefill the audio independently, and the probe score carries render-to-render jitter. AlpacaEval is excluded — it has no binary correctness label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

One absolute threshold, six pools. The deployed gate adapts per pool, but only through label-free quantiles of its own score distribution. To separate what the probe's *ranking* carries from what its *scale* carries, we freeze the single absolute threshold the internal system actually deployed (τ=.628, the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer**: the same τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a fixed cut cannot hold an escalation budget across pools. **Ranking does**: at whatever rate it fires, fixed-τ escalation beats a random reference at the same realized rate on four of five external pools, paired-bootstrap interval excluding zero (Table 10). The exception is Reasoning QA, where τ fires on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free quantile step: it buys budget control, not discrimination.

Does the gate find failures the expert can fix? The probe is fit and read against *local failure*, but what a routing system needs is *expert benefit*: y=1 when the local model is wrong *and* the expert is right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools (Speech TriviaQA .796→.782, Llama Questions .830→.813) and more where the expert is itself weak (Web Questions .773→.689 against a .864 expert ceiling; internal .840→.732). The gate thus remains a usable benefit predictor out of distribution (.63–.81) despite never having seen an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve. On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but only 65.5% of the failures inside the gate's top-scoring 15% are, and that share climbs monotonically as the cut widens (.655/.685/.711 at 15/30/50%) — the gate's most confident failure calls are the ones the expert also misses. Table 11 places the deployed gate against the matched-budget oracle that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time or expert-aware second stage would have to claim.

Table 10: One absolute threshold (τ=.628, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed-τ accuracy minus a random reference at the *same realized rate* (paired bootstrap over ids, B=10^4, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit (y= local wrong $\wedge$ expert right, base rate in the last column).

pool	n	rate	local	acc	rand	Δ 95% CI	$\mathrm{AUC}_{\mathrm{fail}}$	$\mathrm{AUC}_{\mathrm{benefit}}$	base
Speech TriviaQA	250	.284	.664	.832	.750	[+.044, +.121]	.796	.782	.308
Speech Web Questions	250	.360	.568	.720	.675	[+.008, +.084]	.773	.689	.324
Llama Questions	250	.080	.836	.868	.843	[+.006, +.046]	.830	.813	.112
SD-QA	200	.480	.515	.805	.714	[+.045, +.136]	.800	.765	.435
Reasoning QA (zh)	202	.020	.589	.609	.595	[−.001, +.035]	.679	.634	.302
internal frozen test	240	.350	.383	.654	.570	[+.038, +.131]	.840	.732	.533

Table 11: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true y=1 cases first. The gate clears random everywhere; the oracle column is the remaining headroom.

	15% budget			30% budget			50% budget		
pool	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	.710	.760	.816	.755	.836	.964	.816	.912	.972
Speech Web Questions	.612	.640	.720	.657	.700	.868	.716	.764	.892
Llama Questions	.850	.888	.948	.864	.912	.948	.882	.932	.948
SD-QA	.577	.635	.665	.639	.745	.815	.723	.820	.950
Reasoning QA (zh)	.631	.678	.738	.674	.728	.891	.730	.767	.891
internal frozen test	.463	.492	.533	.543	.608	.683	.650	.742	.883

I Transfer latency shapes

Matched-rate precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail .16) the 50% cap is .32 and the deployed probe reaches .304 — 95% of the cap (recall .93). Quoting raw precision would misread the strongest pool as the weakest.

The median left-fold. On easy pools per-arm P50 is non-monotonic (Llama Questions 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe preferentially escalates the queries whose *local* decode is longest — on retrieval pools wrong answers hedge at length (decode time and answer length r=.97), so a wrongness-selecting probe strips the slow tail as a side effect (escalated-at-15% ids: never-arm local P50 1.711 s vs. 1.198 s kept; local accuracy .32 vs. .73). Controls: the probe does not key on length (corr(score, length) $= +.05$; ≈ 0 conditioned on wrongness), and on SD-QA — where wrong answers are short — the same probe selects wrongness identically and produces no fold. The mean is monotonic (1.65→2.06 s); kept-local P50 falls 1.52→0.43 s across tiers (Figure 13). On Reasoning QA the shape inverts usefully: the local chain-of-thought is *spoken*, so every reasoning token enters the response clock, while the expert's chain is hidden behind a flat ∼3.7 s round-trip — escalation truncates the latency tail (P90 13.25→10.94 s) rather than fattening it.

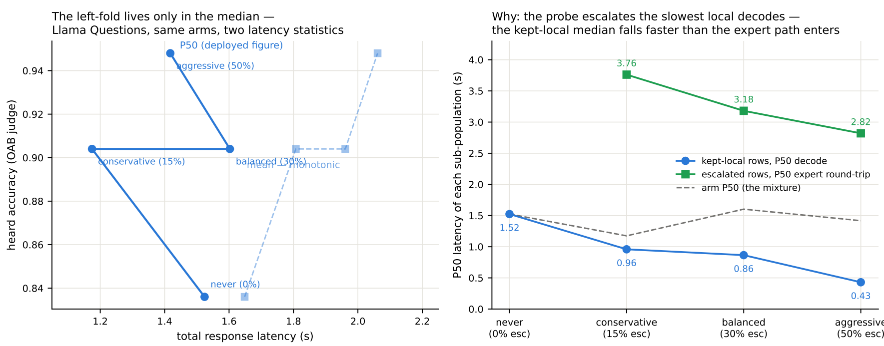


Figure 13: Decomposition of the Llama Questions latency fold: the fold exists only in the median (mean monotonic); kept-local P50 falls 1.52→0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat ∼3 s round-trip.

AlpacaEval scale caveats. The official 4.8 is an offline chat-mode number — our chat-mode control reproduces it at 4.86 — and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop's spoken-reply style plus a token cap, so only live arms are comparable to each other. Selected queries score 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71) because the expert writes better long-form answers — random picks would harvest the same gain.

J Full-pool and per-family figures

Figures 14 (dual-view) on the full frozen pool, and the complete external-family figure set (Appendix K); the noise audit behind the ±.02–.03 floor is Figure 15.

K External-benchmark figure set

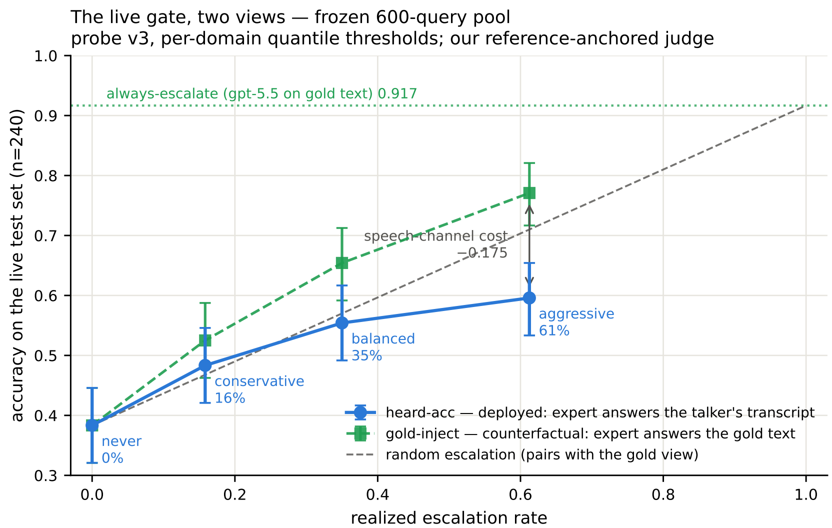


Figure 14: Dual-view live tradeoff, full pool (n=240); companion to Figure 3.

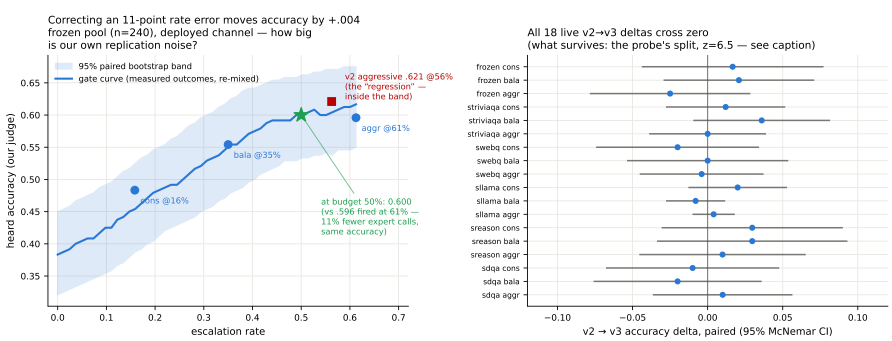


Figure 15: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE .009–.028.

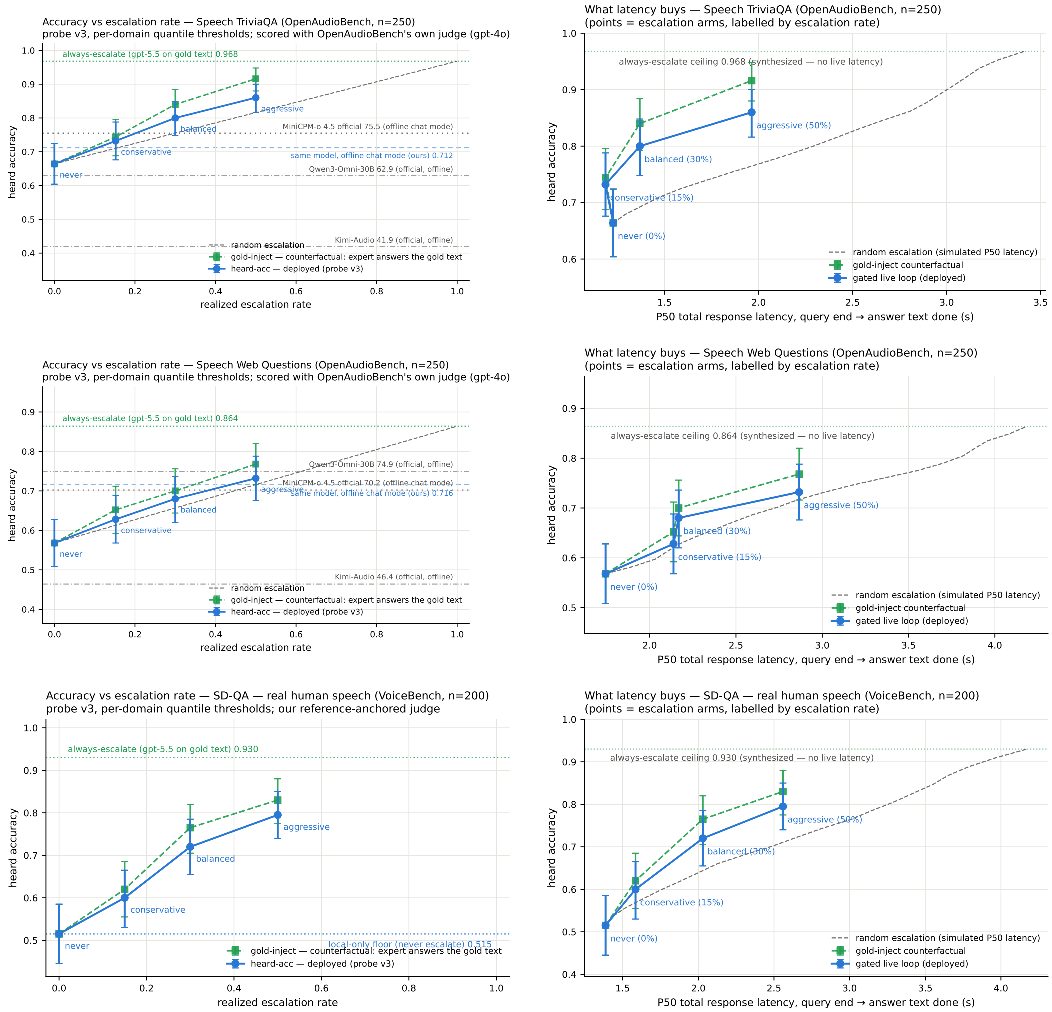


Figure 16: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy–latency (right).

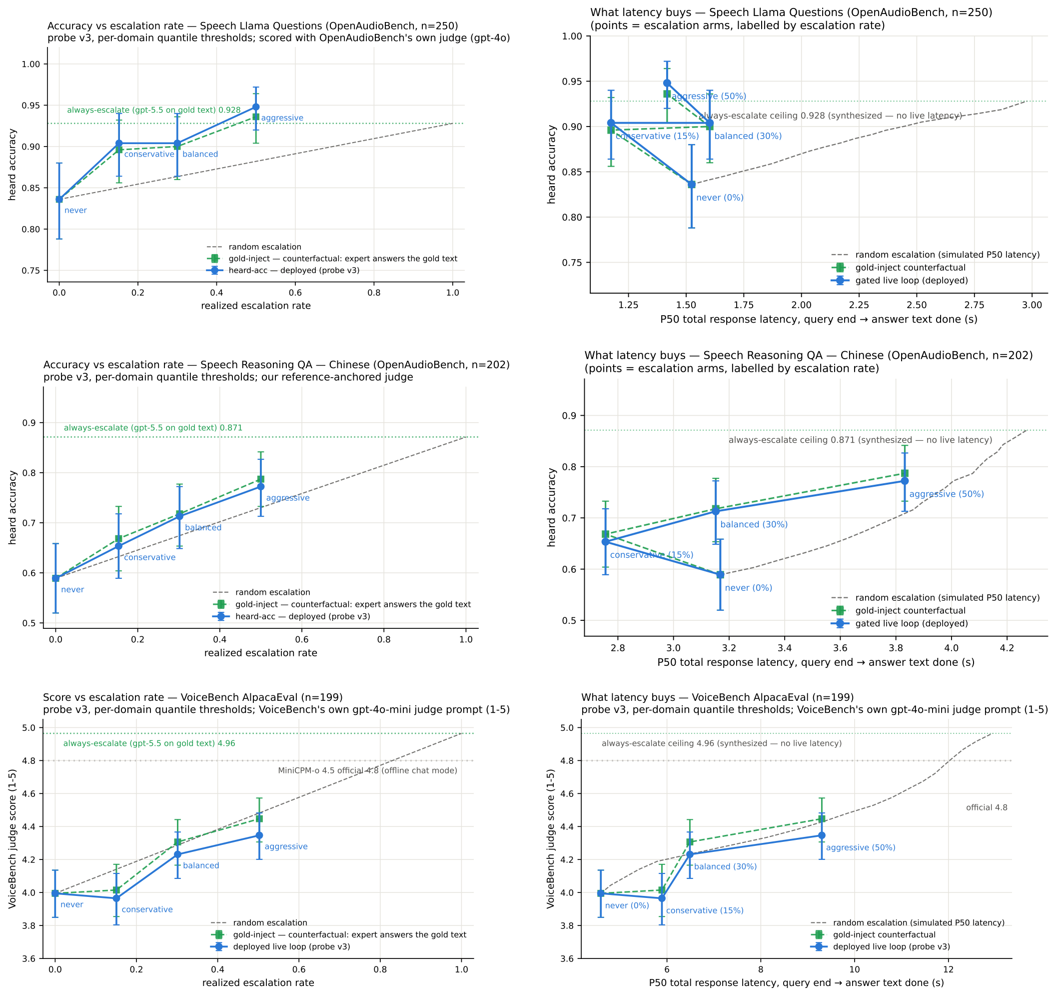


Figure 17: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy–latency (right).

L Judge and self-evaluation prompts

All adequacy labels use the following fixed system prompt (structured outputs with a strict {`reason`, `adequate`} schema; the judge never sees which system produced the answer):

```
You are a strict, fair grader of a small language model's answers.
You are given a QUESTION, an optional REFERENCE answer, and a
CANDIDATE answer produced by a small model.  Decide whether the
CANDIDATE is ADEQUATE, meaning ALL of:  (1) factually correct;
(2) any reasoning shown is valid; (3) it actually and completely
answers the question that was asked.  For multiple-choice questions
the candidate must land on the correct option (by letter or by
unambiguously stating the option's content).  A partially correct,
hedged, refused, off-topic, or truncated answer is NOT adequate.  If
a REFERENCE is provided, grade against it.  Be strict but do not
penalize harmless differences in format or extra correct detail.
```

The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first generated token's distribution:

```
[pre] I am going to show you a question.  Do NOT answer it.  Judge
honestly whether you yourself would answer it correctly.  Question:
{q} Would you answer this question correctly?  Reply with exactly one
word:  Yes or No.
[post] Here is a question and a proposed answer.  Question:  {q}
Proposed answer:  {a} Is the proposed answer correct?  Reply with
exactly one word:  Yes or No.
```

M Reproducibility

All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test split is touched only by the evaluation scripts. Environment pins: torch 2.8.0; transformers 4.51.0 for MiniCPM-o 4.5, with per-model pins for comparison backbones; all experiments run on single-H100 instances. Probe artifacts (weights, feature configuration, per-pool label-free thresholds), the judge prompt above, all trace files, and the per-experiment execution log — including per-run GPU and API spend, on the order of $500 total — ship in the project repository, released upon publication.

TODO (process only — delete before submission)

- **[TODO: P1 — DONE 2026-08-25 (external review applied): main text now ends within 8 pages (refs start mid-page 8). Restructure: 3 contributions + hero figure; merged signal section; live 6pp $\rightarrow$ 1.2pp; router audit 3 lines; title dropped "Zero-Training"; causal claims softened ("consistent with"); no version numbers / artifact filenames in main text; everything cut lives in the appendix (A–L) or git history]**
- **[TODO: P2 — peer + meeting feedback (GPT reviewer pass 2026-08-26, borderline 4/7). (a) and (b) DONE 2026-08-26; (c) dropped as not computable; (d) unchanged. (a) DONE — three claims narrowed: "routing beats scaling the speech backbone" $\rightarrow$ "selective cloud escalation outperforms larger standalone backbones"; "standard readouts fail" $\rightarrow$ "late-layer readouts turn brittle after duplex fine-tuning" (abstract, intro contribution 1, hero + teaser panel-1 titles); and the EOT loop is no longer equated with native full-duplex — intro now carries an explicit scope sentence ("*duplex* names the training regime, not the interaction protocol"), live.tex states the turn-based protocol up front and is retitled "Live End-of-Turn Escalation", and "duplex session/system" $\rightarrow$ "spoken session"/"live system" throughout. (b) DONE — new appendix section (app:fixedthr) + two tables: one absolute threshold ($\tau{=}.628$, frozen internally) applied verbatim to all pools fires at 2–48% yet still beats matched-rate random on 4/5 (ranking transfers, scale does not); expert-benefit oracle (y = local-wrong AND expert-right) with AUC re-scored per pool and a gate/random/oracle frontier at 15/30/50% budgets. Script: scripts/09_fixed_threshold_oracle.py $\rightarrow$ figures/fixed_threshold_oracle.json. Gold-inject view (heard view only observes the $\sim$50% of ids some arm escalated); labels follow each pool's own judge (oab_ok vs adequate — they differ by 13pp on Web Questions, never mix); AlpacaEval excluded, it has no binary label on either side. (c) DONE 2026-08-26 (was DROPPED, then re-run with TTS on): bench_live gained a tts flag (init_tts + Token2wav, canned stall, gen_speak port of demo_app); balanced arm re-run on the frozen 240 ($\sim$50 min H100, expert cache hits). EOT$\rightarrow$first-audible: local p50 .552s/p99 .644s, escalated (canned stall onset) p50 .022s/p99 .029s, expert-content p50 6.4s/p95 26.0/p99 34.0. Same 84/240 fire as text-out; probe read unperturbed (p50 21ms). Landed: live.tex sentence + app:livedetail paragraph. Traces data/frozen_v3tts_traces.jsonl.balanced.shard0-3; analysis scripts/10_tts_latency.py. Fixed en route: fig:timeline-live panel (c) was captioned "first token 0.4 s" but the plotted value is answer_ms = *full* answer decode; caption + figure title corrected and the figure regenerated. (d) PARTIAL DONE 2026-08-26 (was SKIP): duplex-validation sweep ran — the frozen-pool 240 replayed through the DEPLOYED voice loop (talker on, spoken answers) in clean + overlap/barge-in arms via _ws_duplex.py against the live demo; frozen v4 probe, no refit: AUC .871/.856 vs .866 headless (paired deltas inside the replication floor), decision agreement .89/.91, barged 237/239. Landed: live.tex closing paragraph, app:duplexval, intro scope sentence updated, Limitations (vi) narrowed to "concurrent-with-generation read remains open". Data: gate-data volume duplex_sweep/{clean,overlap}.jsonl; analysis _duplex_analyze.py. This also partially recovers (c): the voice loop logs EOT$\rightarrow$first-audible per turn (first_audio_client_ms, p50 $\approx$.58 s, both arms) — usable if a reviewer asks, though only under probe-off replay, not the escalating sweep. Still skipped: human continuity study, second expert, model-level concurrent listen/speak read. Frame the paper as what it is: pre-response competence probing in duplex-trained models + turn-based live escalation, now with a deployed-loop validity check.]**
- **[TODO: P3 — submission day (8-28): delete this section; final anonymization grep (names, rhe9527, modal URLs, gmail); decide public-repo $\rightarrow$ private; OpenReview form + upload (deadline 8-29 AoE, moderation hard limit Fri 8-28)]**